[METRIC]
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SUPERSEDING
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September 15, 1981

COMMERCIAL ITEM DESCRIPTION
GUANIDINE NITRATE

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This CID covers the general requirements for Guanidine Nitrate (GN) used in military explosives.

2. SALIENT CHARACTERISTICS.

2.1 Properties. The GN shall meet the requirements in Table I when tested in accordance with the applicable paragraphs. The quantity shall be as specified in the contract or order (see 6.2).

2.2 Granulation. The GN shall be a white, free flowing material which is free from dirt, oil, grease, or other visible impurity and shall conform to the requirements specified herein. The GN shall have a U.S. Standard Sieve Number of 16 and the quality of material passing through the sieve shall be at least 95% GN by weight.

3. QUALITY ASSURANCE PROVISIONS.

3.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army TACOM-ARDEC, Attn.: AMSTA-AR-QAW-E, Bldg. 12, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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TABLE I. Requirements for guanidine nitrate.

Properties	Requirements	Test Paragraphs
Total Volatiles (Percent (%))	0.8 (Maximum (Max.))	3.4.1
Purity (%)	96.0 (Minimum)	3.4.2
Ammonium Nitrate (%)	1.8 (Max.)	3.4.3
Melamine (%)	0.2 (Max.)	3.4.4
Ash (%)	0.5 (Max.)	3.4.5
Sulfate as Na_2SO_4 (%)	0.1 (Max.)	3.4.6
Free Ammonia (%)	0.5 (Max.)	3.4.7
Water Insolubles (%)	0.2 (Max.)	3.4.8
Temperature Rise (°C)	36 (Max.)	3.4.9

3.2 Granulation. The granulation of the GN shall be determined in accordance with method 204.1 of MIL-STD-650. Free flowing of the GN shall be determined by visual examination when the GN is poured on the nest of sieves.

3.3 Sampling. Two individual representative 200g samples from each batch or lot of the GN shall be obtained as specified in ASTM E300, section 25. Each sample shall be tested independently in accordance to the applicable methods specified in 3.4. If either sample fails to meet any of the test requirements, the batch or lot shall be rejected.

3.4 Test methods. The tests used to determine compliance with the requirements stated in Table I shall be performed on the GN samples obtained in 3.3. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or in accordance with ASTM D2905. Unless otherwise specified herein, all chemicals shall be Reagent Grade or ACS Grade.

3.4.1 Total volatiles. Transfer to a tared glass-stoppered weighing bottle (or a tared aluminum can with a lid) an accurately weighed portion of approximately two grams (g) of the sample. With the stopper off, place the sample, weighing bottle and stopper in an oven at 100°C to 105°C for three hours, remove and cool in a desiccator, and weigh. Calculate the loss in weight as percent total volatiles as follows:

$$\text{Percent total volatiles} = \frac{A-B\ (100)}{A}$$

Where:

A = Weight of specimen before drying (g)
B = Weight of specimen after drying (g)

3.4.2 Purity.

3.4.2.1 Preparation of precipitation/washing solution. Add 7.5g of reagent grade picric acid, 0.075g of reagent grade GN, and 8 milliliters (mL) of concentrated ammonium hydroxide to a 1000mL volumetric flask. Add 500mL of distilled water to the flask and mix thoroughly. Fill the flask with distilled water to the calibration mark and allow to stand for approximately 10 days before use.

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3.4.2.2 Procedure. Weigh 0.2g of reagent grade GN into a 250mL flask. Add 15mL distilled water and 5 to 10 drops concentrated ammonium hydroxide. Heat the flask and contents on a hot plate until the GN dissolves. Remove the flask from the hot plate and allow it to cool to room temperature and filter through filter paper. To the filtrate slowly add 100mL of the precipitation solution prepared in 3.4.2.1 to the GN solution while continually swirling the flask. Allow the solution to stand 8 to 12 hours at room temperature, then filter through a porcelain filtering crucible, of fine porosity, which is attached to a vacuum system containing a crucible holder and filtering flask. Rinse the flask with portions of the precipitating/washing solution. Wash the precipitate with the washings in the flask. Evacuate the crucible to dryness. Dry the filtering crucible and contents to a constant weight in a drying oven at 105°C + 5°C. Cool the filtering crucible and contents in a desiccator and weigh. Calculate the value of the GN to guanidine picrate factor (A) as follows:

$$A = \frac{C\ (D)}{100\ (B)}$$

Where:

A = GN to guanidine picrate factor
B = Weight of precipitate in crucible (g)
C = Weight of sample used (g)
D = Purity of GN reagent in percent (%)

3.4.2.3 GN purity determination. Repeat the procedure described in 3.4.2.2 using the GN sample instead of the reagent grade GN. Calculate the percent GN as follows:

$$\text{Percent GN} = \frac{E\ (A)\ (100)}{W}$$

Where:

E = Weight of precipitate in crucible (g)
W = Weight of sample used (g)
A = GN to guanidine picrate factor determined in 3.4.2.2

3.4.3 Ammonium nitrate.

3.4.3.1 Preparation of the formaldehyde solution. Add a few drops of phenolphthalein to a 25mL of formaldehyde. Slowly titrate the solution with 0.5N or 0.33N sodium hydroxide to a faint pink color.

3.4.3.2 Procedure. Add an accurately weighed portion of the sample, approximately l0g, to a 400mL beaker containing 200mL water. Heat the contents of the beaker on a hot plate and stir the solution until all the GN dissolves. To the beaker and contents add 5.0mL of the standardized formaldehyde, then 2 to 3 drops of phenolphthalein solution. Titrate the solution with 0.5N or 0.33N sodium hydroxide to a faint pink color. Calculate the percent ammonium nitrate as follows:

$$\text{Percent ammonium nitrate} = \frac{A\ (N)\ (M)\ (100)}{1000\ (W)}$$

Where:

A = mL of sodium hydroxide used in titration
N = Normality (N) of sodium hydroxide
M = Molecular weight of ammonium nitrate
W = Weight of GN sample (g)

3.4.4 Melamine. To a 250mL volumetric flask, containing 200mL hot water (70°C to 80°C), quantitatively transfer an accurately weighed portion of the sample,

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approximately l0g. Mix the contents of the flask until sample dissolves completely. Allow the contents of the flask to cool to room temperature and fill the flask to the mark with distilled water and mix. Pipette a 50mL portion of the solution into a 250mL beaker and perform a potentiometric titration on the solution using 1N sulfuric acid and a calomel/antimony electrode. The potential shift occurs at a pH of approximately 3.5 (190 to 200 millivolts). Calculate the percent melamine as follows:

$$\text{Percent melamine} = \frac{A\ (N)\ (M)\ (100)}{100\ (P)}$$

Where:

A = mL of sulfuric acid used in potentiometric determination
N = Normality (N) of the sulfuric acid
M = Molecular weight of melamine (126)
P = Weight in grams of GN introduced in the potentiometric determination. This value may be calculated as follows:

$$P = \frac{A\ (B)}{C}$$

Where:

A = Weight of GN dissolved (g)
B = Volume of solution used in analysis (mL)
C = Capacity of volumetric flask (mL)
P = 2.0 when sample size, flask size, and solution volume drawn with pipette are exactly as detailed in the procedure

3.4.5 Ash. Weigh accurately approximately a 5g portion of the GN sample into a tared porcelain crucible which was properly cleaned and dried. Carefully and slowly heat the crucible until sample decomposes. Continue heating in an electric furnace (900°C ± 50°C) until constant weight is obtained. Calculate the percentage of ash using the following formula:

$$\text{Percent ash} = \frac{A\ (100)}{W}$$

Where:

A = Weight of contents after ignition (g)
W = Weight of GN sample (g)

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3.4.6 Sulfate (as sodium sulfate).

3.4.6.1 Procedure. Transfer an accurately weighed portion of the GN sample, approximately 5.0g, to a 400mL beaker containing 200mL of hot water (80°C to 90°C). Use a glass stirrer rod and stir to dissolve the sample. If necessary, filter the solution through the filter paper. Acidify the solution by adding 1.0mL of the concentrated hydrochloric acid. Then add 1.0mL of barium chloride solution (10% by weight). Boil the resultant solution for approximately five minutes on a hot plate, remove from the hot plate, allow the solution to cool to room temperature with occasional gentle stirring. Allow the beaker and contents to remain at room temperature for a minimum of 10 hours. Then, heat to boiling on a hot plate and quantitatively filter the contents of the beaker through a previously ignited and tared fine porosity fritted glass crucible using warm 20mL portions of distilled water at 50°C ± 5°C to aid in the transfer. Wash the crucible and precipitate with 20mL portions of warm water. Draw each portion of the warm water through before adding the next 20mL portion of warm water. Discontinue the washings when the filtrate shows no chlorides when tested with silver nitrate solution (1% by weight). Place the crucible and contents for one hour in an oven maintained at 100°C ± 5°C. Then place the crucible and contents in a muffle furnace (600°C ± 25°C) for one hour. Remove crucible and contents from furnace, cool in a desiccator and weigh. Calculate the percent sodium sulfate as follows:

$$\text{Percent sodium sulfate} = \frac{60.86\ (A)}{W}$$

Where:

A = Weight of residue (g)
W = Weight of sample (g)

3.4.7 Free ammonia.

3.4.7.1 Procedure. Accurately weigh approximately a l.0g portion of the sample into a 250mL beaker. Add approximately l00mL of distilled water and then add four drops of methyl red indicator solution (0.1% by weight in isopropyl alcohol). If the solution turns yellow, using magnetic stirring, titrate the solution with 0.33N HCl until a red color end point is obtained. Record the number of mL of HCl as A.

Calculate the free ammonia as:

$$\text{Percent free ammonia} = \frac{1.703\ (A)\ (N \text{ of HCl})}{\text{weight of sample}}$$

3.4.8 Water insolubles.

3.4.8.1 Procedure. Transfer an accurately weighed portion of the sample, approximately 10g, to a 800mL beaker containing 400mL of warm distilled water (50°C to 75°C). Stir to dissolve sample. Quantitatively transfer the contents of the beaker through a tared, fine porosity sintered glass crucible. Wash the crucible and contents with five 20mL portions of warm water. Allow each portion of the wash water to go through the crucible before adding the next portion. Dry the crucible and contents in an oven, maintained at 105°C ± 5°C for two hours, then cool in a desiccator and weigh. Calculate the water insolubles as follows:

$$\text{Percent water insoluble impurities} = \frac{A\text{-}B\ (100)}{W}$$

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Where:

A = Final weight of crucible (g)
B = Initial (tare) weight of crucible (g)
W = Weight of sample (g)

3.4.9 Temperature rise.

3.4.9.1 Procedure. Set up a nitration rack over a container of cold distilled water into which the reaction mixture can be dumped and quenched with cold distilled water if the temperature rise is excessive. Pour 100mL of 98% technical grade sulfuric acid into a 500mL flask; allow to stand for two minutes, then record the temperature (NOTE: The temperature should be between 25°C and 30°C.) Add a weighed portion of 18g ± 0.1g of the GN sample, using a triple beam balance (of at least 2000g capacity), into the flask. At the end of two minutes, record the temperature; then dispose of the mixture in the approved waste container. Calculate the temperature rise by subtracting the initial temperature from the final temperature. A duplicate determination should be made.

4. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

5. NOTES.

5.1 Source of documents.

5.1.1 The Code of Federal Regulations (CFR) is available by mail order from the Superintendent of Documents, ATTN: New Order, PO Box 371954, Pittsburgh, PA 15250-7954.

5.1.2 ASTM standards are available from the American Society of Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

5.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Standards and Specifications (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Quantity required (see 2.1).
d. Packaging and marking requirements (see 4).

5.3 Lot formation. An indefinite quantity of GN produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of GN that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

5.4 Material Safety Data Sheets. Products meeting the requirements of this CID may be subject to the requirements of 29 CFR 1910.1200 and FED-STD-313 for Material Safety Data Sheets (MSDS).

5.5 Transportation issues. Products meeting the requirements of this CID may be subject to the requirements of 49 CFR 100-185 for transportation of explosives and other dangerous articles.

5.6 Precautionary information. Precaution: This CID covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that

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all applicable safety rules, regulations, and procedures must be followed in handling and processing these materials.

5.7 Keywords.
Ammonia
Ammonia hydroxide
Ammonium nitrate
Barium chloride
Chemicals
Explosives
Formaldehyde
Hydrochloric acid
Isopropyl alcohol
Phenolphthalein
Picric acid
Sodium hydroxide
Sodium sulfate
Silver nitrate
Sulfuric acid

MILITARY INTEREST:

Custodian
Army-AR

Review Activity
DLA-GS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing Activity:
Army-AR

Project 6810-1603

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-59339	**2. DOCUMENT DATE (YYMMDD)** 980915

Guanidine Nitrate

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Mr, Gary Vander Sande	b. TELEPHONE *Include Area Code)* (1) Commercial (973) 724-2163 (2) AUTOVON 880-2163
Commander, U.S. Army TACOM-ARDEC ATTN: AMSTA-AR-QAW-E, Bldg 12 Picatinny Arsenal, NJ 07806-5000	Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 (703) 756-2340 DSN: 289-2340

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